

HINTS & SOLUTION

Q.1 (B)

Let the mass of sample = 100 g

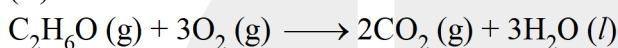
$$\text{Mass of nitrogen in the sample} = 42 \text{ g} = \frac{42}{14} = 3 \text{ mole}$$

$$\therefore \text{Mass of pure urea} = \frac{3}{2} \times 60 = 90 \text{ g}$$

$$\therefore \text{Purity} = 90\%$$

$$\left[\begin{array}{cc} \text{Urea} & \text{N} \\ \therefore \frac{1 \text{ mole}}{2 \text{ mole}} & \\ \frac{3}{2} \text{ mole} \leftarrow & 3 \text{ mole} \end{array} \right]$$

Q.2 (D)



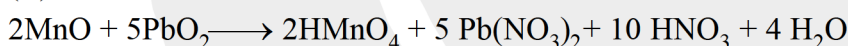
$$\begin{array}{ccc} 20 & 80 & 0 \\ 0 & 80-60 & 40 \end{array}$$

$$V_i = 20 + 80 = 100 \quad V_f = 20 + 40 = 60$$

$$\text{reduction in volume} = V_i - V_f = 100 - 60 = 40$$

$$\% \text{ Reduction in volume} = \frac{40}{100} \times 100 = 40\%$$

Q.3 (A)



Q.4 (B)

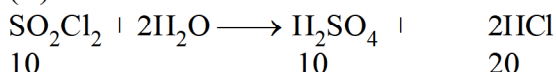
Assume we have 1 mole sample which contains X_A mole of solute.

$$\therefore \text{Weight of solvent in 1 mole solution} = (1 - X_A) \cdot 17$$

$$\therefore 17(1 - X_A) \text{ gm solvent contain } X_A \text{ mole solute}$$

$$\therefore 1000 \text{ gm solvent contain } \frac{X_A \times 1000}{17(1 - X_A)} \text{ mole solute} = \frac{58.8 \times A}{(1 - X_A)}$$

Q.5 (A)



$$\begin{array}{ccc} 10 & 10 & 20 \end{array}$$

$$\text{Number of moles of SO}_2\text{Cl}_2 = 50 \times 0.2 = 10 \text{ millimoles}$$

$$\therefore \text{Number of millimoles of H}_2\text{SO}_4 = 10$$

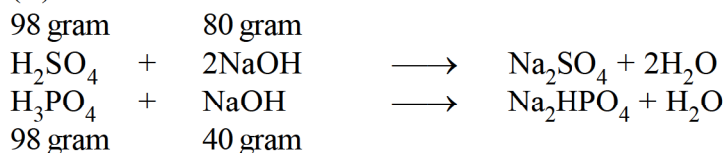
$$\therefore \text{Number of millimoles of HCl} = 20$$

$$\therefore \text{Number of millimoles of H}^+ = 20 + 20 = 40$$

$$\therefore \text{Number of moles of Ba(OH)}_2 = 20 \text{ millimoles}$$

$$\text{Now } V = \frac{20}{0.2} = 100 \text{ ml}$$

Q.6 (C)



40 gram NaOH reacts with 49 gram H_2SO_4 & 98 gram H_3PO_4 to form sulphate & dihydrogen phosphate.

wt. ratio



1 : 2 Ans.

Q.7 (B)

Let the wt. of sample = x g

$$\text{Mass of silica} = x \times \frac{50}{100} = 0.5x$$

$$\text{Mass of water} = x \times \frac{15}{100} = 0.15x$$

$$\text{Mass of water left} = (0.15x - 10) \text{ g}$$

$$\text{New mass of sample} = (x - 10) \text{ g}$$

$$\% \text{ of silica} = \frac{0.5x}{(x - 10)} \times 100 = 52 \quad x = 260 \text{ g}$$

$$\% \text{ of water} = \frac{0.5x - 10}{(260 - 10)} \times 100 = 11.6\%$$

Alternative :

Let the mass of sample = 100 g

Mass of silica = 50 g, Mass of water = 15 g

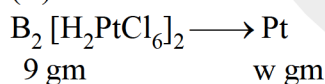
Mass of all other components except water = $100 - 15 = 85 \text{ g}$

Since, on heating only water is removed hence composition of all other components will remain same

$$\frac{50}{85} = \frac{52}{(100 - x)} \quad (x \Rightarrow \% \text{ of water in final sample})$$

$$x = 11.6$$

Q.8 (C)



POAC for Pt :

$$\frac{9 \times 2}{2 \times 175 + 410 \times 2} = \frac{w}{195}$$

or, $w = 3 \text{ gm}$ Ans.

Q.9 (A)

Q.10 (A)

$$10 \times 20 = 200$$

$$4 \times 480 = 1920$$

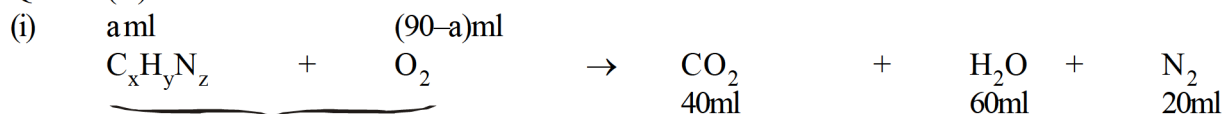
$$\Rightarrow 2120 \text{ gm Ans.}$$



Q.11 (C)

Q.12 (B)

Q.13 (A)



Applying POAC for C

$$a \times x = 40 \quad \dots(i)$$

$$\text{for H } y \times a = 120 \quad \dots(ii)$$

$$\text{for N } z \times a = 2 \times 20 \quad \dots(iii)$$

$$\text{for O } 2 \times (90-a) = 2 \times 40 + 60$$

$$90 - a = 70$$

$$a = 20 \text{ ml}$$

(ii) Volume of $\text{O}_2 = 70 \text{ ml}$

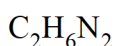
$$\text{Mole \% of O}_2 = \text{volume \% of O}_2 = \frac{70}{90} \times 100 = \frac{700}{9} \%$$

(iii) Solving (i), (ii), (iii)

$$x = 2$$

$$y = 6$$

$$z = 2$$



Q.14 (A) (B) (C) (D)

Using phenolphthalein,

$$1 \times M_{\text{Na}_2\text{CO}_3} \times 10 = 2.5 \times \frac{1}{5} = 0.5$$

$$M_{\text{Na}_2\text{CO}_3} = 0.05 \text{ M}$$

Using methyl orange

$$2 \times M_{\text{Na}_2\text{CO}_3} \times 10 + 1 \times M_{\text{NaHCO}_3} \times 10 = 7.5 \times \frac{1}{5}$$

$$2 \times 0.05 \times 10 + 10 M_{\text{NaHCO}_3} = 1.5$$

$$M_{\text{NaHCO}_3} = 0.05 \text{ M}$$

$$\text{Mass of } M_{\text{Na}_2\text{CO}_3} \text{ in 10 ml solution} = 10 \times 0.05 \times 106 \text{ gm} \times 10^{-3} = 0.053 \text{ gm}$$

$$\text{Mass of NaHCO}_3 \text{ in 10 ml solution} = 10 \times 0.05 \times 84 \times 10^{-3} \text{ gm} = 0.042 \text{ gm}$$

Q.15 (A) S, (B) R, (C) Q, (D) P

%Mole

$$(A) \quad \% \text{ of } (a-1) = \frac{a-(a+4)}{(a+4)-(a-1)} \times 100 = \frac{4}{5} \times 100 = 80\%$$

$$(B) \quad \% \text{ of } a = \frac{(2a-5a)}{(5a-a)} \times 100 = \frac{3a}{4a} \times 100 = 75\%$$

$$(C) \quad \% \text{ of } (a+1) = \frac{(a+2)-(a+3)}{(a+3)-(a+1)} \times 100 = \frac{1}{2} \times 100 = 50\%$$

$$(D) \quad \% \text{ of } (a-1) = \frac{a-(a+2)}{(a+2)-(a-1)} \times 100 = \frac{2}{3} \times 100 = 66.67\%$$

%Mass

$$(A) \quad \frac{80 \times (a-1)}{a} = \frac{80a-80}{a}$$

$$(B) \quad \frac{75 \times a}{2a} = \frac{75a}{2a}$$

$$(C) \quad \frac{50 \times (a+1)}{a+2} = \frac{5a+50}{a+2}$$

$$(D) \quad \frac{66.67 \times (a-1)}{a} = \frac{66.67a-66.67}{a}$$

Q.16 0.66

$$X_{\text{CH}_3\text{OH}} = 0.1$$

$$\frac{40}{9} \text{ moles in } 1000 \text{ gm} ;$$

$$\frac{X}{\frac{40}{9} + \frac{1000}{18} + x} = 0.1$$

$$x = 0.1 \left[\frac{40}{9} + \frac{1000}{18} + x \right] ;$$

$$10x = x + \frac{40}{9} + \frac{1000}{18}$$

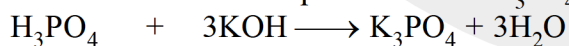
$$9x = \frac{80+1000}{18} ;$$

$$x = \frac{1080}{18 \times 9}$$

$$\frac{n_{\text{C}_2\text{H}_5\text{OH}}}{n_{\text{CH}_3\text{OH}}} = \frac{40}{9} \times \frac{18 \times 9}{1080} = 0.66 \text{ Ans.}$$

Q.17 10 ml

Maximum volume is required in case of H_3PO_4

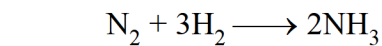


$$\frac{1.96}{98} = 0.2 \quad 0.06$$

$$56 \times 0.06 = V \times \frac{33.6}{100}$$

$$V = 10 \text{ ml} \quad \text{Ans.}$$

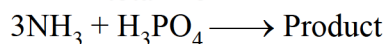
Q.18 6.8



$$t = 0 \quad 1 \quad 3$$

$$t = \text{final} \quad 1 - x \quad 3 - 3x \quad 2x$$

$$\text{total mol} = 4 - 2x$$



$$\text{Mole of H}_3\text{PO}_4 = M \times V$$

$$\Rightarrow 2.5 \times 0.2 = 0.5 \text{ mol}$$

$$\text{for complete neutralization mol of NH}_3 = 0.5 \times 3 = 1.5 \text{ mol}$$

$$\text{so that } 2x = 1.5$$

$$x = 0.75 \text{ mol}$$

$$\text{It means total mole in I}^{\text{st}} \text{ reaction} = 4 - 2x$$

$$= 4 - 2 \times 0.75$$

$$= 4 - 1.5 = 2.5 \text{ mol}$$

$$\bar{M} = \frac{\text{total mass}}{\text{total mol}} = \frac{28 + 6}{2.5} = 13.6$$

$$\text{V.D.} = \frac{\bar{M} \text{ wt.}}{2} = \frac{13.6}{2} = 6.8$$

Q.19 (a) 0.836, (b) 103.103

(a) Molality of $\text{SO}_3 = 2$

1000 gm of H_2SO_4 contains = 2 mole of SO_3

$$n_{\text{H}_2\text{SO}_4} = \frac{1000}{98}$$

$$n_{\text{SO}_3} = 2$$

$$X_{\text{H}_2\text{SO}_4} = \frac{\frac{1000}{98}}{\frac{1000}{98} + 2} = 0.836$$

(b) 1160 gm mixture contains 2 mol SO_3

1160 gm can react with 36 gm H_2O

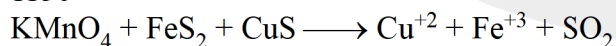
$$100 \text{ gm can react with } \frac{36}{1160} \times 100 = 3.103$$

$$\% \text{ labelling} = 100 + 3.103 = 103.103$$

Q.20 12.5 l

Q.21 6 l

Q.22 1150



$$n = 11 \quad n = 6$$

$$\text{Eq. of KMnO}_4 = \text{Eq. of FeS}_2 + \text{Eq. of CuS}$$

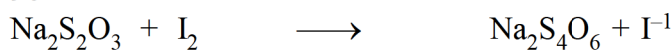
$$\frac{N \times V}{1000} = \frac{MV_{\text{FeS}_2}}{1000} \times n + \frac{MV_{\text{CuS}}}{1000} \times n$$

$$\frac{N \times 20}{1000} = \frac{10 \times 1}{1000} \times 11 + \frac{20 \times 1 \times 6}{1000}$$

$$N = \frac{110 + 120}{20} = \frac{230}{20} = 11.5 \text{ N Ans.}$$



Q.23 90

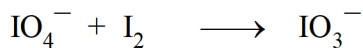


$$n_f = 1 \quad n_f = 2$$

$$0.1 \times 1 = 2 \times \text{mole of I}_2$$

$$\text{mole of I}_2 = \frac{0.1}{2} \text{ moles}$$

$$\text{milli moles} = 0.1 \times 1000 / 2 = 50$$



$$n_f = 2 \quad n_f = 10$$

$$10 \times \text{mole of I}_2 = 2 \times 0.2 \times 1$$

$$\text{mole of I}_2 = 0.04$$

$$\text{milli mole} = 40$$

$$\text{total milli mole} = 40 + 50 = 90$$

Q.24 0002

$$100 \text{ ml} \quad 11.2 \text{ V H}_2\text{O}_2 \text{ (Molarity 1 M)}$$

$$200 \text{ ml} \quad 2.8 \text{ m}$$

$$\rho = 1.0952 \text{ gm/ml}$$

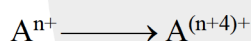
$$\text{Molarity} = \frac{2.8}{1095.2/1.0952} \times 1000 = 2.8 \text{ M}$$

$$\text{Final molarity } M_1V_1 + M_2V_2 = M_f[V_1 + V_2] \times \frac{110}{100}$$

$$100 \times 1 + 2.8 \times 200 = M_f \times 300 \times \frac{110}{100}$$

$$M_f = 2 \text{ M}$$

Q.25 40



$$t = 0 \quad a \quad 0$$

T.R. $t = t \quad a - x \quad x$

Let at both time 10 ml is titrated

$$t = 0, \quad a \times 10 \times 2 = 30 \times N$$

Let Normality of titrating reagent = N

$$a = \frac{30}{10 \times 2} \times N = \frac{N}{10 \times 2} \times 30$$

$$\text{At } t = 10 \text{ min, } (a - x) \times 10 \times 2 + x \times 10 \times 5 = 45 \times N$$

$$\text{or } a \times 10 \times 2 - x \times 10 \times 2 + x \times 10 \times 5 = 45 N$$

$$\text{or } 30 x = 15 N$$

$$x = \frac{15}{30} N = \frac{1}{2} N$$

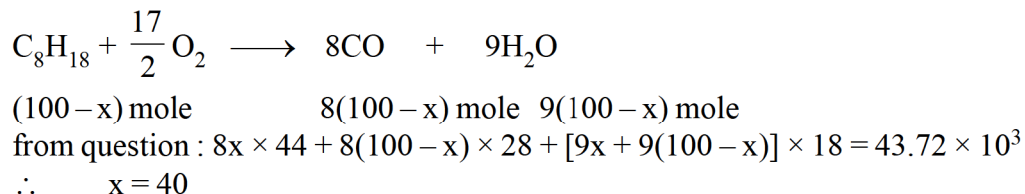
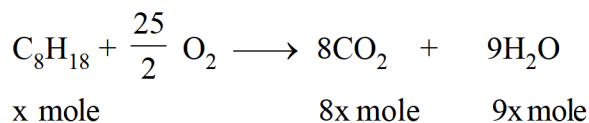
$$K = \frac{1}{t} \ln \frac{a}{(a - x)} = \frac{1}{10} \ln \frac{3/2 N}{3/2 N - 1/2 N}$$

$$= \frac{1}{10} \ln \frac{3/2}{1} = \frac{1}{10} \times 0.4 = 0.04$$



Q.26 0040

$$\text{Mole of octane taken} = \frac{10 \times 1000 \times 1.14}{114} = 100$$



Q.27 0004

11.2L at 0°C and 1 atm of compound $\Rightarrow \frac{1}{2}$ mole 'X' is always ≥ 15.5 g

$\therefore$ In 1 mole, of compound of 'X' ≥ 31 g

$\therefore$ Atomic weight of 'X' = 31

($\because$ in one mole of compound of any element there should always be 1 mole of atoms of that element)

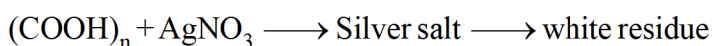
Also 11.2 L of vapour of 'X' weighs $\Rightarrow 62$ g

(At 0°C, 1 atm) $\left(\frac{1}{2} \text{ mole} \right)$

$\therefore$ Molecular weight of 'X' = 124

$$\text{Atomicity of 'X'} = \frac{124}{31} = 4$$

Q.28 0134



Amount of residue obtained (W_2) = 54 gm

No. of moles of silver salt used (n_1) = 0.25

$$\frac{W_2}{108 \times n} = n_1$$

$$\frac{54}{108 \times n} = 0.25 \quad n = 2$$

$$m_{(\text{COONa})_2} = 2 \times 12 + 4 \times 16 + 2 \times 23 = 134 \text{ gm/mole}$$

Q.29 0.0623M

Q.30 1.5, 40 ppm, pH = 2.6989